Latency audit and 14-second test plan

17 October 2026. Functional checks pass. The 14-second target is NOT verified on an RTX 4090.

The 4090 was unavailable during this audit. The machine actually tested was an i9-14900K, 32 GB RAM and RTX 3080 10 GB, driver 617.14. The intended machine is a 14th-generation i9 (exact SKU unconfirmed), 64 GB RAM and RTX 4090 24 GB. Montenegrin and the four assigned cloned voices remain requirements. The primary package remains Gemma 4 12B Q8; QAT Q4 is an explicitly optional dialogue comparison.

What was run

196 automated service/helper tests passed, including speech-cache identity, language settings, model selection, reply budgets and interruption behavior.

Real dialogue models, not estimated token rates

Four sequential Alex conversation turns per configuration used the same questions and seed 42. Speech, transcription, game rendering and playback were excluded. The first turn includes uncached prompt work; the other three form the warm median. Models can produce different-length answers, so this is a pipeline comparison, not a controlled model-quality study or a pure hardware speed ratio.

Dialogue configuration on the RTX 3080	First turn	Warm median	Trials
Gemma Q8, GPU-layer setting 32	26.063 s	21.495 s	4
Gemma Q8, GPU-layer setting 36	38.286 s	21.413 s	4
Gemma QAT Q4, all eligible layers on GPU	4.127 s	3.486 s	4

Q8 dialogue decoded at about 10-11 tokens/s; fully offloaded QAT Q4 commonly decoded around 68-70 tokens/s. **3.486 seconds is dialogue preparation only.** It is not a 4090 result and does not include the voice pipeline.

Real BF16 cloned speech

Each voice was warmed, then measured in two baseline/optimized pairs with the order alternated. Eight of eight paired WAV hashes matched exactly. The optional adapter keeps the existing sampler, reference, precision, seeds and stopping rules; it removes per-frame CPU copies and repeated position-tensor allocation.

Voice / test emotion	Baseline median	Adapter median
Alex / anxious	3.634 s	3.627 s
Ivan / angry	2.885 s	2.893 s
Stefan / sad (pack's neutral fallback)	3.875 s	3.757 s
Nikola / sad	4.550 s	4.350 s

These short clips lasted approximately 2.9-4.9 seconds. The gain is small and varies; Ivan was slightly slower. It cannot explain or solve a 23-second wait by itself. Four further paired checks of the final adapter with optional unused-VRAM release also matched WAV hashes. This is sample parity, not proof for every text. The silence/activity checks passed; a native listener must still assess diction, identity and emotional delivery. Whole-clip versus sentence delivery can change phrasing even when precision and voice identity inputs stay the same.

Transcription and the complete service path

Multilingual large-v3-turbo ran on CPU in INT8. Four known Montenegrin clips were decoded with beam sizes 5 and 1. Median times were approximately 3.07 and 2.98 s. After punctuation/case normalization, three of four transcripts exactly matched the intended words with either beam size. Both decoded Ivan's “traže” as “otraže”. This does not distinguish a speech-generation error from an ASR error. Beam 1 is an optional speed/recognition comparison, not a large demonstrated gain.

The real STT -> appraisal -> dialogue -> BF16 speech -> Rhubarb service path also completed for all four characters. Because this 10 GB card cannot accommodate the target's combined full-GPU workload, dialogue was deliberately placed on CPU for this functional check. First prepared speech took 52.981-84.685 s; median 58.880 s. Transcription took 2.95-3.18 s. During the run, Windows reported about 1,806 MiB shared and 9,019 MiB dedicated GPU memory for the speech process. These observations show a constrained configuration; they are not a prediction of 4090 latency. Shared allocation alone does not prove which tensors were paged or establish causation.

The existing Windows player separately passed its actual HTTP/WAV playback, pause/resume and interruption regression using exact approved recording fixtures (`STREAM_CHECK_OK`, exit 0). This was a batch-mode desktop integration check, not an AI speed test, physical speaker-output measurement or headset/frame-time test.

Why the reported 23-30 seconds are plausible

1. **The Q8 model is only partly on GPU.** This GGUF has 48 transformer blocks. In bundled llama.cpp b10909, GPU-layer counts also include the output layer: setting 36 selects 35 blocks plus output, leaving 13 blocks on CPU. Each generated token still waits for those CPU blocks. A low overall CPU percentage does not eliminate a bottleneck in the eight inference threads or memory bandwidth.
2. **Two model requests run sequentially.** The first appraises the student's remark and updates the relationship; the second writes the validated reply. JSON control fields also consume generated tokens. At 10 tokens/s, 140 tokens alone require roughly 14 seconds, before prompt processing, speech recognition or synthesis.
3. **Sentence mode is not waveform streaming.** The complete first sentence WAV and its lip alignment must be ready before it is published. Later sentences render serially while the game plays earlier ones. If rendering the next sentence takes longer than the current clip's playback, a gap is unavoidable in this architecture.
4. **Memory must cover all workloads and their peaks.** Q8, BF16 speech, both CUDA contexts, KV caches, prefill/codec buffers and the game share the card. Reported idle VRAM of 20.8 GB does not prove paging, but does not guarantee peak headroom. More GPU layers or larger batches can make a tight configuration slower.
5. **Retries can multiply latency.** An over-budget or wrong-language dialogue reply is regenerated. Failed speech activity checks can also regenerate audio. A low token ceiling that truncates JSON or speech is not a valid solution.
6. **Recorded references are conditioning, not ready answers.** Exact approved answers and generated WAV cache hits are fast, but a new reply still needs neural synthesis. The small exact-line library cannot accelerate arbitrary new wording.

RAM use of 35 GB on a 64 GB PC is not, by itself, evidence of exhausted RAM. Idle CPU/GPU utilization being low is normal for resident services waiting for requests. The user's 4090 shared-memory use, clocks, active processes, actual token rates and stage timings remain unknown until that machine can be measured.

Changes made in this update

- Fixed batch, mouth-animation and MTP toggles silently forcing the selected model back to Q8. Unsupported MTP now fails without changing the selected configuration.
- Added the optional `latency-bf16` profile and complete matching example config: QAT Q4 fully offloaded, BF16 assigned voices, sentence delivery, an 18-word ordinary reply budget, compact directions, cached/stable prompts, Flash Attention, Q8 KV, and CPU Whisper beam 1. Explicit allowed requests for detail retain their larger disclosure budget. Replies are rewritten when necessary, never cut mid-speech.
- Added a clear per-turn word target below the ceiling to reduce unnecessary length rewrites. An overly strict ceiling can itself add retries; inspect metrics. A real eight-turn preset check respected all applicable budgets. The extra target removed the length rewrites observed in three ordinary turns in the first pass. Detail-capable open questions can also receive the larger budget; the ordinary ceiling does not apply to every question. Detailed replies request a short, meaningful first sentence before the fuller explanation.
- Added experimental `q8-gpu-bf16`: preserves Q8 plus BF16, tries complete GPU offload, uses one 4K context instead of two, smaller 128/64 batches, sentence delivery and releases unused speech allocator memory after synthesis. **Fit/speed on 24 GB are unverified.** Releasing allocator blocks can add allocation work to later calls.
- Added configuration-based sentence override and restoration of each voice pack's own mode, without editing reference recordings, recipes or manifests.

- Added optional transfer reduction in project-owned code; pinned model files stay unchanged. Speech-cache keys include the new generation settings and adapter code.
- Made Whisper beam size and compute type effective settings; beam 1 also uses best-of 1. Regional transcription remains `sr`, Latin output, no translation.
- Added audio-envelope mouth movement as an optional way to bypass Rhubarb's wait; it trades phoneme precision for faster readiness. Rhubarb can be restored separately.
- Added per-segment synthesis/alignment/readiness timings, inference timestamps, predicted later gaps, failures in benchmark reports and a configurable 14-second preparation target. Diagnostics export only allowlisted timings, not transcripts.
- Added `Check GPU Memory.cmd` and reproducible four-voice decoder comparison tooling.

The update changes Python services, settings and diagnostics. The shipped player protocol is compatible; the fixture regression above exercised it. No new Unity PCM streaming player or optimized SGLang speech server is included.

Install and test on the 4090

1. Close the game and run **Stop Services.cmd**. Merge the new latency-update ZIP's `PsychologyVR` contents into the existing application folder. Keep your original `services/config.local.json`; the patch does not replace it. Settings changes save backups under `services/.runtime/config-backups`.
2. To preserve Q8 and BF16 first, try **Optimization Settings.cmd -> 13**. It is an experiment. Start the desktop game, inspect VRAM/shared-memory behavior during generation, and reject it if it fails to load, spills or causes pauses. The 4090 has not been tested with this setting. Use option 1 to return to partial-Q8 BF16.
3. The strongest measured dialogue alternative is **PC Settings.cmd -> 6**, which installs/selects Gemma QAT Q4 (about 7 GB extra download if absent), followed by **Optimization Settings.cmd -> 8**. This restores BF16 explicitly and sets the Montenegrin low-latency options. QAT Q4 changes dialogue precision; compare quality.
4. Keep Rhubarb first. If alignment is material, compare option **9** (audio envelope) and restore phoneme alignment with **10**. Sentence override is **11**, restoration of the pack's mode is **12**. Options 4-7 now preserve your selected dialogue model.
5. Run **Benchmark Responses.cmd**. For recorded microphone input, use: `powershell -File tools/benchmark-response.ps1 -Turns 20 -Audio path\to\mono.wav`. This measures transcription through prepared WAV, not audible playback. Inspect `services/.runtime/response-benchmark/report.json`, especially segment waits, cache sources, retries, errors and maximum latency, not just its median.
6. In the actual game, measure at least 20 new turns across all four voices. Include cold/warm turns, ordinary questions, explicit detail and changing emotions. Measure from microphone release/end of speaking to meaningful audible speech, and measure later sentence gaps separately. Record whether speech was generated or cached. Repeat desktop and VR separately. A preparation result below 14 seconds cannot certify release-to-audible latency; a cache hit cannot certify new-text performance.

The acceptance goal is **<=14 seconds to meaningful speech**, correct Montenegrin, assigned voice, no dropped words, reliable interruption and acceptable rendering. Longer content can continue speaking beyond 14 seconds; its start and later gaps are distinct from total generation time. Do not label a filler sound or cached generic acknowledgement as the completed answer.

If the 4090 still misses the goal with both models fitting comfortably, measure the slow stage before changing more settings. A different streaming speech backend with bounded cache/concurrency, CUDA graphs and a Unity PCM receiver is a larger next implementation, requiring platform and voice-quality validation. It has not been installed or benchmarked here. A second GPU or separate speech host can retain Q8 and BF16 if one card cannot fit them; it is not an automatic latency guarantee.

Mobile notifications and remaining limits

The user confirmed an iPhone, connection through ChatGPT Codex/Remote, and iOS notifications enabled. Setup: desktop Settings -> Connections -> Control this PC, scan the QR code, use the same account/workspace, then enable ChatGPT notifications in iOS Settings. Keep the host awake and online. Task completion alerts can open the relevant task. Delivery is still checked by receiving a real alert; this audit has not independently observed the phone receiving one. No SMS service was configured.

The repository remains private. The original upstream-owner write invitation to IgorCulafic remains pending. Upstream's latest release is still v0.5.0-metahuman-cast; the audit did not find a newer release to replace the downloaded base.

Conclusion: functional regressions checked here pass, and full GPU dialogue offload is the clearest measured improvement. Neither these local component tests nor the constrained 3080 pipeline prove the 4090 meets 14 seconds. The missing final step is the real 4090 test with the intended voices, game and microphone.

Evidence and sources

Settings reference

Setting	Original Q8/BF16 example	Optional low-latency profile 8
Dialogue weights	Gemma Q8	Gemma QAT Q4
GPU-layer setting	32	99 (all eligible layers)
Context / slots	4096 per slot / 2	4096 per slot / 2
Batch / microbatch	256 / 128	256 / 128
Inference CPU threads	8	8
Flash Attention / KV	On / Q8	On / Q8
MTP	Off	Off
Ordinary spoken ceiling	Relationship budget	18; shorter target; detail exception
Speech weights	BF16	BF16
Speech reference routing	Assigned cast	Same assigned cast
Delivery mode	Voice pack recipe	Sentence override
Transfer-reduction adapter	Off	On; small variable gain
STT / device	large-v3-turbo / CPU	Same multilingual model / CPU
STT beam size	Default 5	1; recognition comparison
Language	cnr dialogue, sr STT, Latin	Same; no English fallback
Lip alignment	Rhubarb	Preserves selection; option 9 skips it

Experimental profile 13 instead retains Q8, requests complete GPU offload, changes batch/microbatch to 128/64 and cache slots to one, and enables unused-VRAM release. The per-slot context remains 4096. These settings reduce buffers/cache demand but do not prove that Q8, BF16 and the game fit within the 4090's usable memory budget.

- `docs/benchmarks/2026-10-17-latency-audit.json`: summarized local measurements.
- Pinned llama.cpp offload allocation: the output layer is included in this build's offload count.
- Upstream release.
- Official mobile setup.
- Official Remote notifications.
- Earlier optimization roadmap remains background; this audit supersedes its untested assumptions and updates the target to 14 seconds.